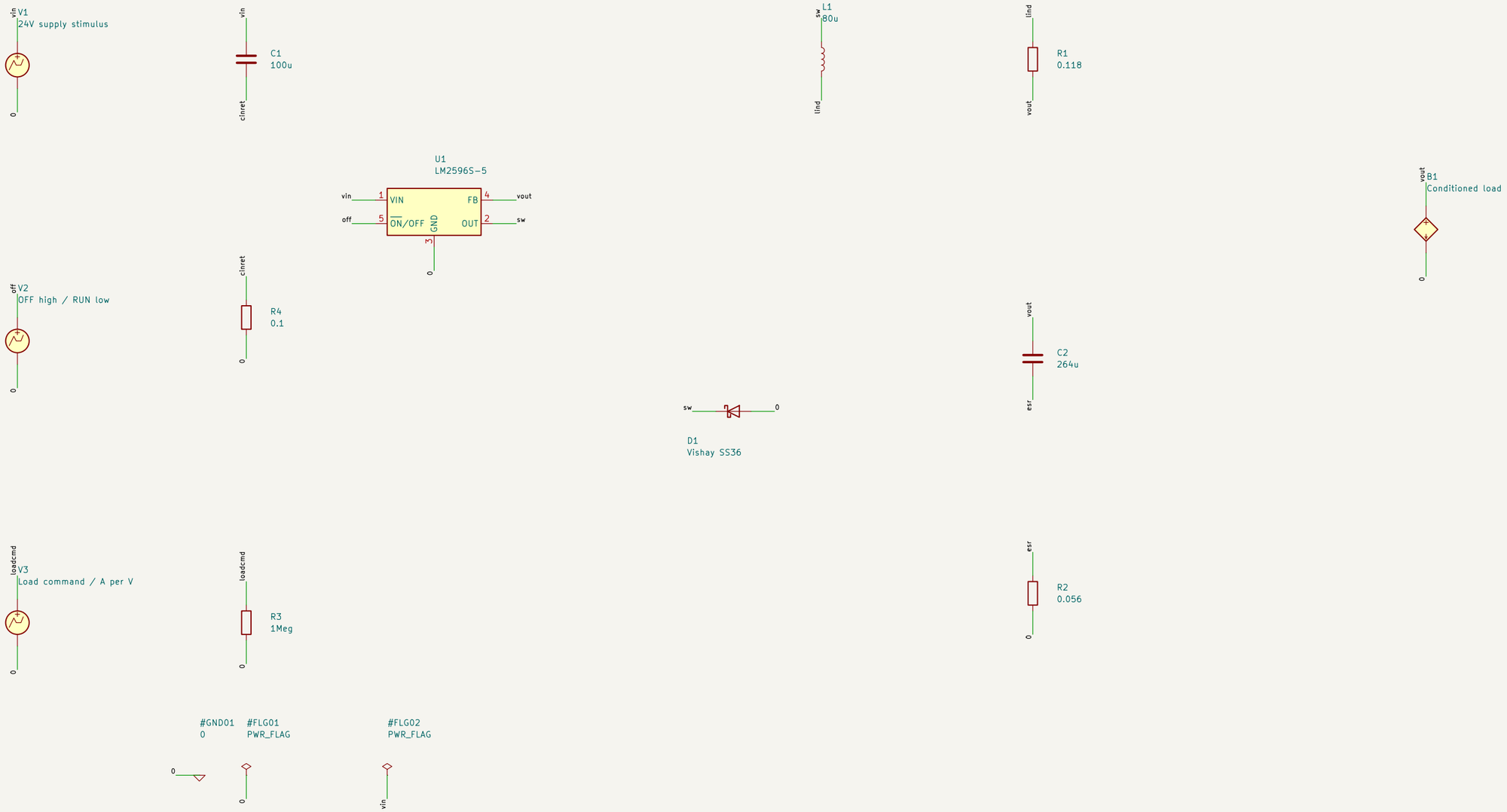
KICAD / NGSPICE — LM2596 FIXED 5V — ESR_LOWER_BOUND

Converter-only simulation study. Not installed or routed on PCB B.
TI regulator and Vishay SS36 models are unmodified. Final component and thermal qualification remain required.



U1 node order VIN FB OUT GND ON_OFF_N = physical pins 1 4 2 3 5. FB directly senses VOUT.
R1 = inductor DCR; R4 = input ESR; R2 = lumped output series resistance. Intended hardware adds 56mOhm.
Stress ESR cases may violate that hardware bound. V3 commands a load, reduced below 4.75V.

```
.include "../../vendor_model/LM2596_5P0_TRANS.LIB"

.include "../../vendor_model/ss36.llb"

.options method=gear maxord=2 reltol=0.003 abstol=1n vntol=1u

.tran 100n 12m 0 100n
```

Sheet: /		
File: lm2596_5v_esr_lower_bound.kicad_sch		
Title: LM2596S-5 KiCad/ngspice simulation — esr_lower_bound		
Size: A3	Date: 2026-11-05	**Rev: 0.2 SIMULATION**
KiCad E.D.A. 10.0.6		Id: 1/1